MIL-M-28787/189C(EL)
21 November 1980
SUPERSEDING
MIL-M-28787/189B(EC)
12 June 1978

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

PROCESSOR, ERROR, S/D, ANALOG, KEY CODE SHX

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope This document establishes the requirements and test procedures for the procurement of an electronic S/D error processor module.

1.2 Classification This specification covers one class of modules.

Class 1 as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.
MIL-STD-1605 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirments . Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.93 failures per million operating hours when predicted in accordance with MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Naval Electronic Systems Command, ATTN ELEX 5043, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class 1 as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1388 and as specified herein.

3.5.1 Key code. The module key code shall be SHA.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/189-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for power supply voltage (V_{CC}) of 5.0 ±0.5 Vdc and ±15 ±1.5 Vdc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

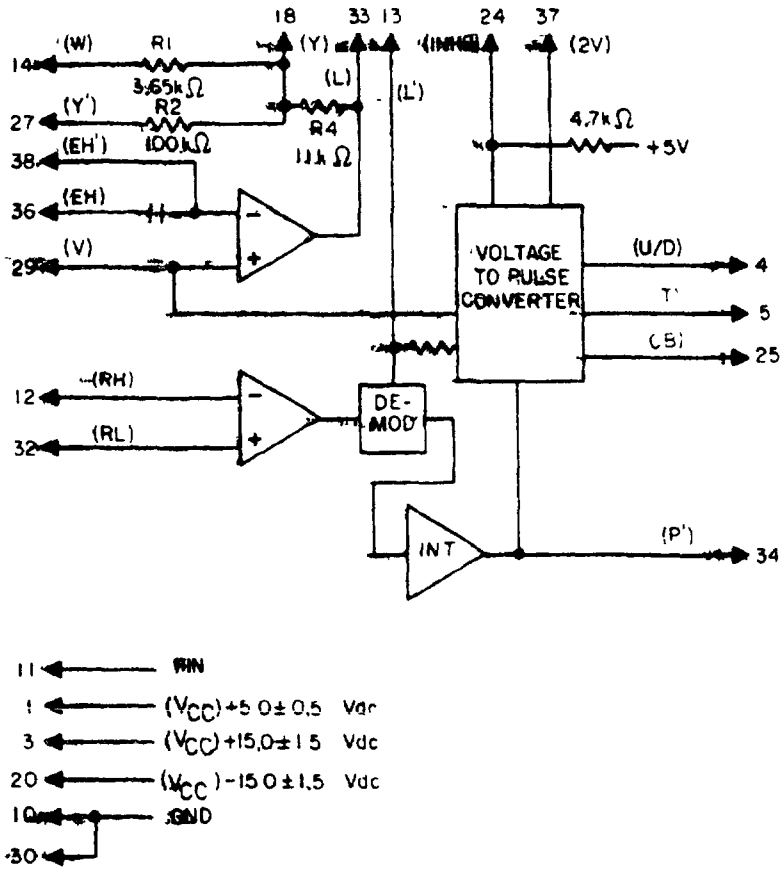
4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be noninductive and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity (SEMP-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through G shall be performed in any order with the following exceptions. If test conditions A, B, C, D, E, F, or G are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, F, G where applicable, followed by the remaining tests in the sequence.

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NOTE The following are unused pins 2, 6, 7, 8, 9, 15, 16, 17, 19, 21, 22, 23, 26, 28, 31, 35, 39, and 40

FIGURE 1 SHX module function diagram, S/D error processor.

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1-A19	Isolation	II	S	10	∞	1.0	∞	10	∞	1 0	∞	MΩ
B	Input breakdown current (digital)	III	S	0	1	0	1	0	1	0	1	mA
C1,C3,C5	Hot socket	IV	Q	64	250	64	250	64	250	64	250	kHz
C2,C4,C6	Hot socket	IV	Q	0	0	0	0	0	0	0	0	Hz
D	V stress	V	Q	← See table V and 4/ →								
E	Vs stress	VI	Q	← See table VI and 4/ →								
F	Module stress	VII	Q	← See table VII and 4/ →								
G1-G3	Output short circuit current	VIII	S	← See table VIII and 4/ →								
G4	Output short circuit current	VIII	S	15	30	15	30	15	30	15	30	mVrms
H1,H2	Maximum count	IX	X	64	250	64	250	64	250	64	250	kHz
H3,H4	Maximum count	IX	Q	60	250	60	250	60	250	60	250	kHz
J1-J3	Scaling resistance	X	X	See table X and 4/								
J4	Continuity (pin 11 to pin)	X	S	0	1	0	1	0	1	0	1	Ω
J5	Continuity (pin 10 to pin 30)	X	Q	0.0	0.1	0 0	0.1	0.0	0.1	0 0	0.1	Ω
J6	Continuity (pin 10 to pin 30)	X	Q	0	1	0	1	0	1	0	1	Ω
K1	Capacitance	XI	S	11.25	18.75	11 25	18.75	11 25	18.75	11.25	18.75	µF
K2-K4	Power supply capacitance	XI	Q	0 075	0.125	0 075	0.125	0.075	0.125	0.075	0 125	µF
L1	Input current (digital), low level	XII	X	1.0	1 4	1.0	1.4	1.0	1.4	1.0	1.4	mA
L2	Input current (digital), high level	XII	X	300	440	300	440	300	440	300	440	µA
M1,M2	Input current (analog), V	XIII	X	-1 0	1.5	-1.0	1.5	-1 0	1.5	-1 0	1 5	mA
M3	Input current (analog), 2 V	XIII	X	0	1	0	1	0	1	0	1	mA
M4	Transfer function (opamp)	XIII	X	4.80	5 20	4.80	5.20	4 80	5.20	4.80	5 20	Vrms
M5	DC offset	XIII	X	-25	25	-25	25	-25	25	-25	25	mV
N1-N3	Power supply current	XIV	S	← See table XIV and 4/ →								
N4-N6	Power supply current	XIV	Q	← See table XIV and 4/ →								
P1-P12	Dynamic test	XV	S	← See table XV and 4/ →								

See footnotes at end of table.

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TABLE I Module electrical requirements - Continued

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25° ±5°C ambient temperature				Operating temperature 0° to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
P13-P18	Dynamic test	XV	X	←See table XV and 4/→								
P19-P22	Dynamic test	XV	S	←See table XV and 4/→								
P23,P24	Dynamic test	XV	X	←See table XV and 4/→								
P25	Dynamic test	XV	S	←See table XV and 4/→								
Q1,Q2	Inhibit delay	XVI	X	0 0	2.5	0 0	2.5	0.0	2.5	0.0	2.5	µs
R1,R3	Input impedance check	XVII	S	0	5	0	5	0	5	0	5	Vrms
R2,R4	Input impedance check	XVII	S	-20	20	-20	20	-20	20	-20	20	mVdc
S1-S6	Gain	XVIII	X	←See table XVIII and 4/→								
f1,f3	Voltage controlled oscillator (2 V)	XIX	S	4	40	4	40	4	40	4	40	kHz
f2,f4	Voltage controlled oscillator (5 V)	XIX	S	10	105	10	105	10	105	10	105	kHz
U1-U4	Null zone	XX	X	97.6	720.0	97.6	720.0	97.6	720 0	97.6	720.0	kΩ
U5	Null zone calculation	XX	X	1.3	3.0	1.3	3.0	1.3	3.0	1.3	3.0	--

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ The initial and end-of-life limits for acceptance requirements and operating temperature are the same.

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4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through U in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions H through U shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 3. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests H through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests H through U shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6 NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

* 6.3 Cross reference of classification. Part number M28787/189-1 supersedes part number M28787/189. The two parts are interchangeable.

6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I.

Maximum power dissipation	1.1 W (outputs not loaded)
Input voltage	
Low level	-0.5 V minimum, 0.8 V maximum
High level	3.5 V minimum, V_{CC} maximum

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

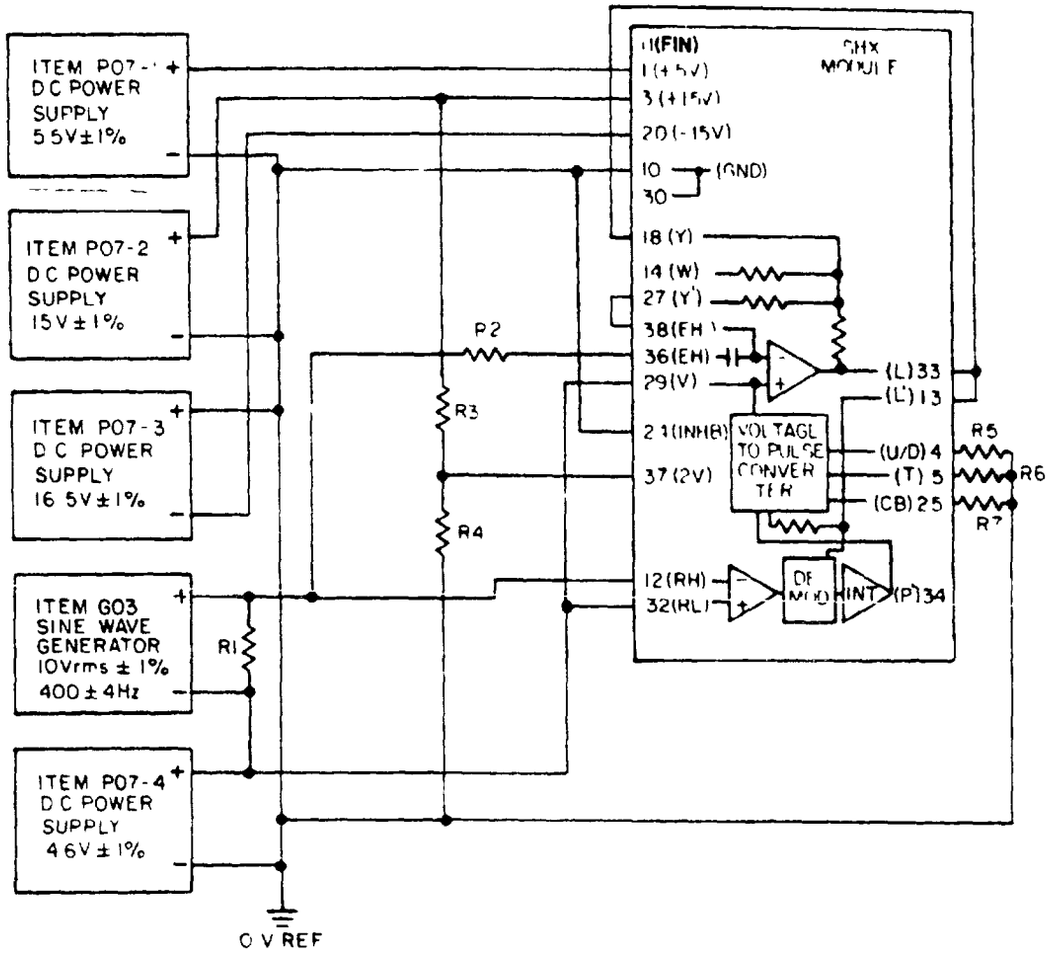
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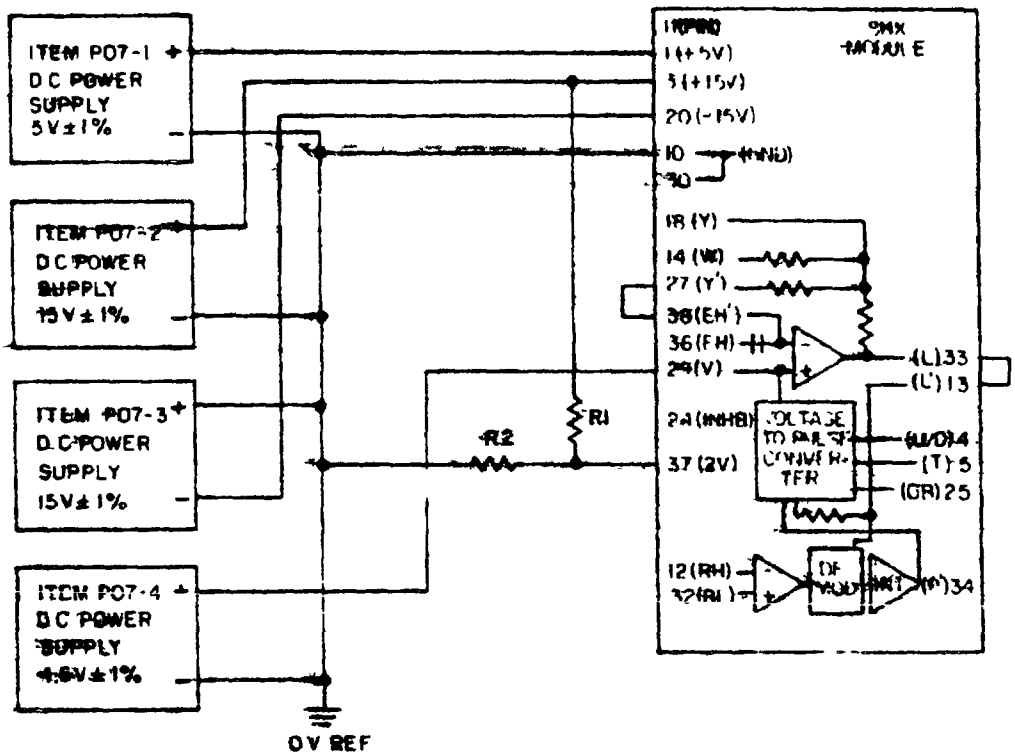


NOTES

1 R1 = 600 Ω ± 5%, 1 W
2 R2 = 5.0 kΩ ± 5%, 1/4 W
3 R3 = 9.09 kΩ ± 1%, 1/8 W
4 R4 = 6.19 kΩ ± 1%, 1/8 W
5 R5-R7 = 1 kΩ ± 1%, 1/8 W

FIGURE 2 Operating temperature, component temperature, and life-test circuit

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NOTES:

1 R1 = 9.99 kΩ ±1%, 1/8 W
2 R2 = 6.19 kΩ ±1%, 1/8 W

FIGURE 3. Vibration test circuit

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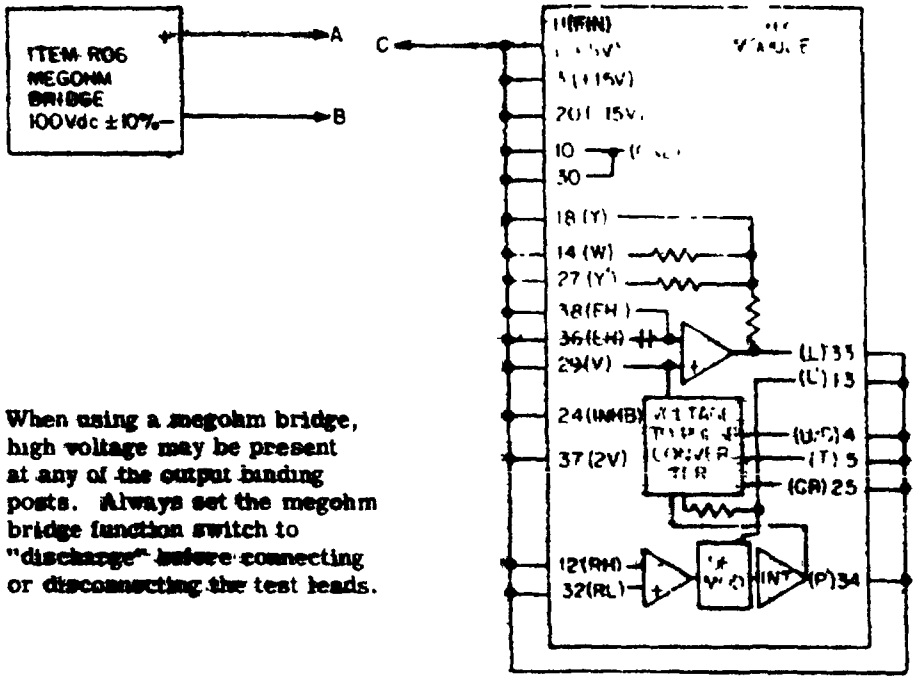
TABLE II Isolation test

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
A1	Isolation, pins to fin pin	11	C,2,6,7,8,9,15,16,17,19,21, 22,23,26,28,31,35,39,40	10	∞	MΩ	4
A2	Isolation, used to unused pins	C	2,6,7,8,9,15,16,17,19,21,22, 23,26,28,31,35,39,40				
A3	Isolation, unused to unused pins	40	2,6,7,8,9,15,16,17,19,21,22, 23,26,28,31,35,39				
A4		39	2,6,7,8,9,15,16,17,19,21,22, 23,26,28,31,35				
A5		35	2,6,7,8,9,15,16,17,19,21,22, 23,26,28,31				
A6		31	2,6,7,8,9,15,16,17,19,21,22, 23,26,28				
A7		28	2,6,7,8,9,15,16,17,19,21,22, 23,26				
A8		26	2,6,7,8,9,15,16,17,19,21,22, 23				
A9		23	2,6,7,8,9,15,16,17,19,21,22				
A10		22	2,6,7,8,9,15,16,17,19,21				
A11		21	2,6,7,8,9,15,16,17,19				
A12		19	2,6,7 8,9,15,16,17				
A13		17	2,6,7 8,9,15,16				
A14		16	2,6,7 8,9,15				
A15		15	2,6,7 8,9				
A16		9	2,6,7,8				
A17		8	2,6,7				
A18		7	2 6				
A19		6	2				

TABLE III. Input breakdown current (digital) test

Test condition	Parameter	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
B	Input breakdown current (digital)	0	1	mA	5

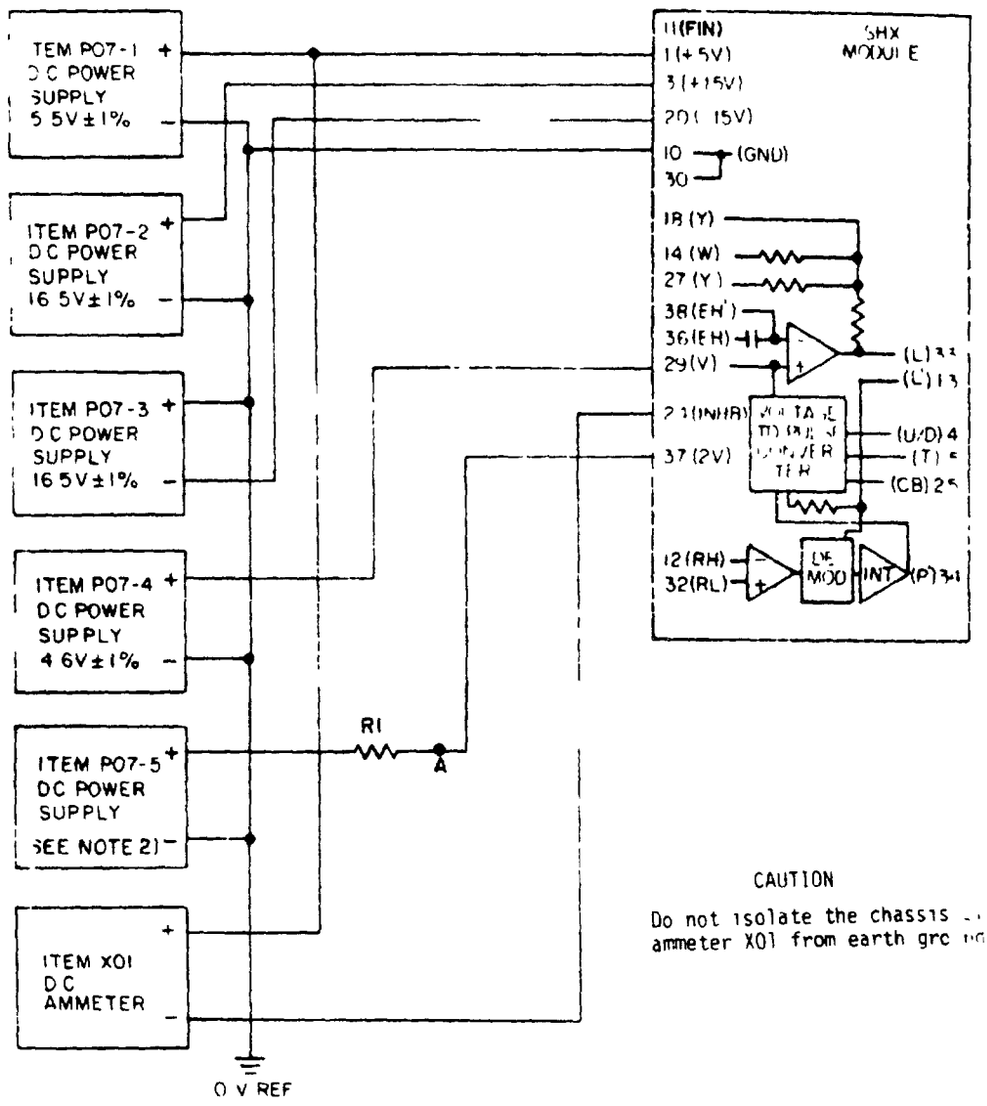
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WARNING: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

FIGURE 4. Isolation test circuit for table II.

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NOTES
1 R1 = 240Ω ±1%, 1/8 W
2 Adjust item P07-5 for 6 2 V ±1% at point A

FIGURE 5 Input breakdown current (digital) test circuit for table III

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TABLE IV Hot socket test

Test condition	Parameter	Position switch S4	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
C1	Hot socket	1	64	250	kHz	6
C2		2	0	0	Hz	
C3		1	64	250	kHz	
C4		2	0	0	Hz	
C5		1	64	250	kHz	
C6		2	0	0	Hz	

TABLE V. V stress test

Test condition	Parameter	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
D	V stress 1/	← Survival →			8

1/ Satisfaction of test condition D is demonstrated by acceptance completion of remaining test conditions in the order specified in 4.4.2.

TABLE VI. Vs stress test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
E	Vs stress 1/	← Survival →			9

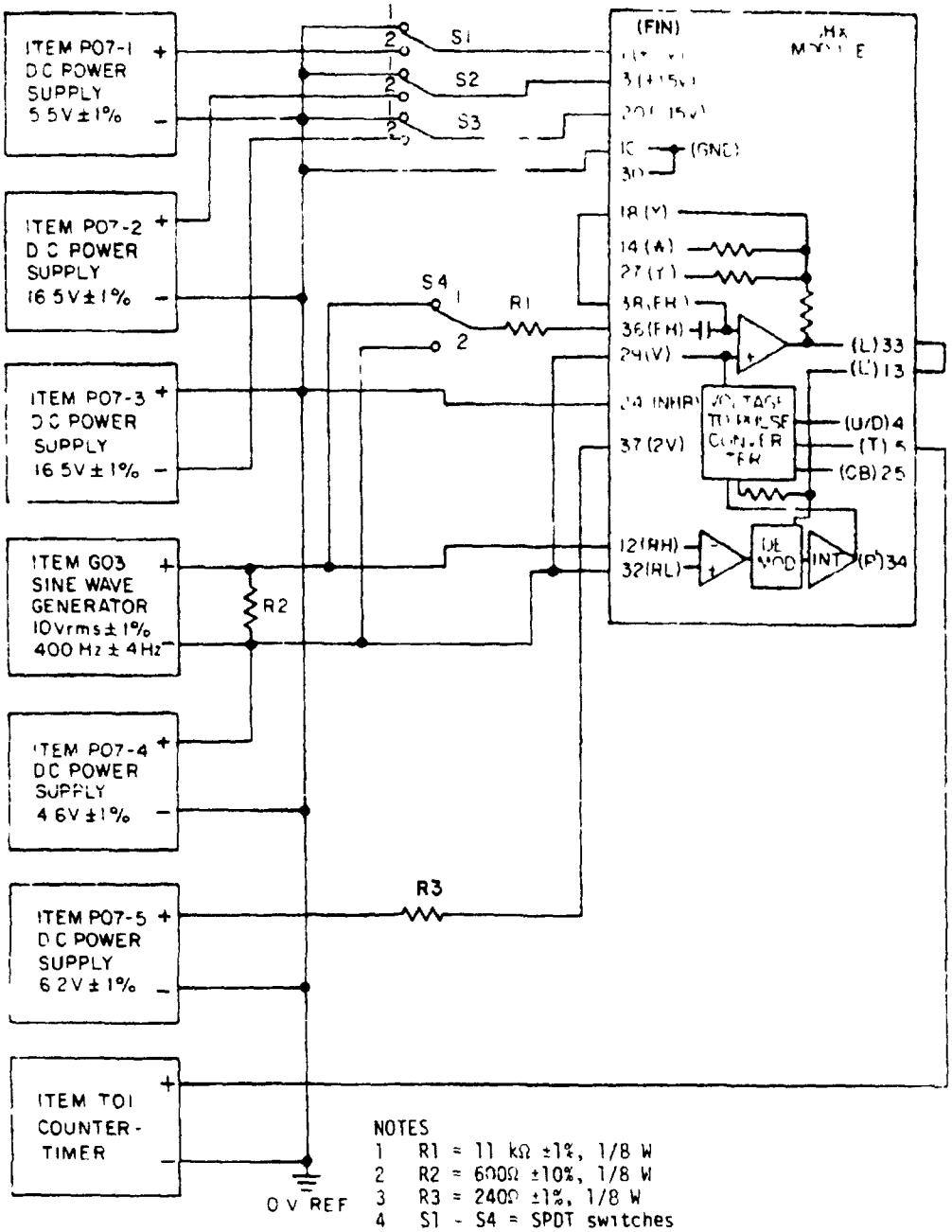
1/ Satisfaction of test condition E is demonstrated by acceptance completion of remaining test conditions in the order specified in 4.4.2

TABLE VII. Module stress test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
F	Module stress 1/	← Survival →			10

1/ Satisfaction of test condition F is demonstrated by acceptance completion of remaining test conditions in the order specified in 4.4.2

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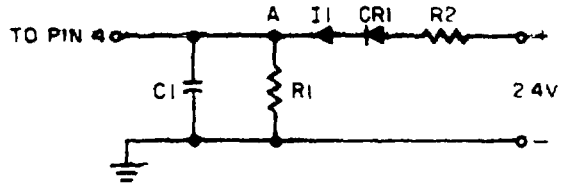


NOTES
1. R1 = 11 kΩ ±1%, 1/8 W
2. R2 = 600Ω ±10%, 1/8 W
3. R3 = 240Ω ±1%, 1/8 W
4. S1 - S4 = SPDT switches
5. Switching sequence
   (a) Position switches S1 - S4 to position 1. Switch S1 to position 2, S2 to position 2, then S3 to position 2. Perform test conditions C1 and C2.
   (b) Reposition switches S1 - S3 to position 1. Switch S2 to position 2, S3 to position 2, then S1 to position 2. Perform test conditions C3 and C4.
   (c) Reposition switches S1 - S3 to position 1. Switch S3 to position 2, S1 to position 2, then S2 to position 2. Perform test conditions C5 and C6.
6. Wait a minimum of 1 ms after S1 is in position before item T01 is gated on for test conditions C1 - C6.
7. Load pins 4, 5, and 25 as specified on figure 7.

* FIGURE 6. Hot socket test circuit for table IV.

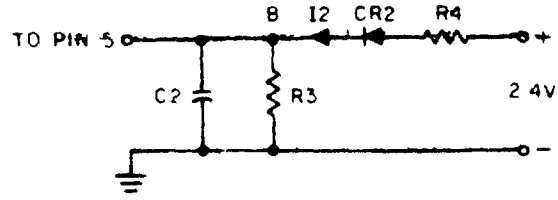
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1. Load for U/D line pin 4·



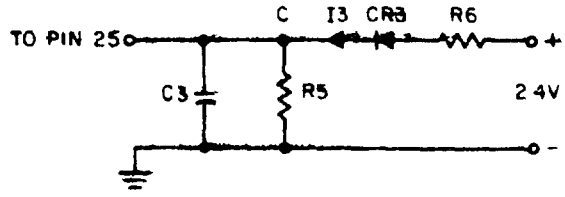
(a) $R1 = \frac{2.4V}{250\ \mu A} = 9.6\ k\Omega \pm 1\%,\ 1/8\ W$
(b) CR1 = 1N4150, or equivalent.
(c) Select R2 such that I1 = 8.0 mA ± 1% when point A is held at 0.5 V.
(d) C1 = 100 pF ± 5%.

2 Load for T line pin 5



(a) $R3 = \frac{2.4V}{250\ \mu A} = 9.6\ k\Omega \pm 1\%,\ 1/8\ W$
(b) CR2 = 1N4150, or equivalent
(c) Select R4 such that I2 = 8.0 mA ± 1% when point B is held at 0.5 V
(d) C2 = 100 pF ± 5%

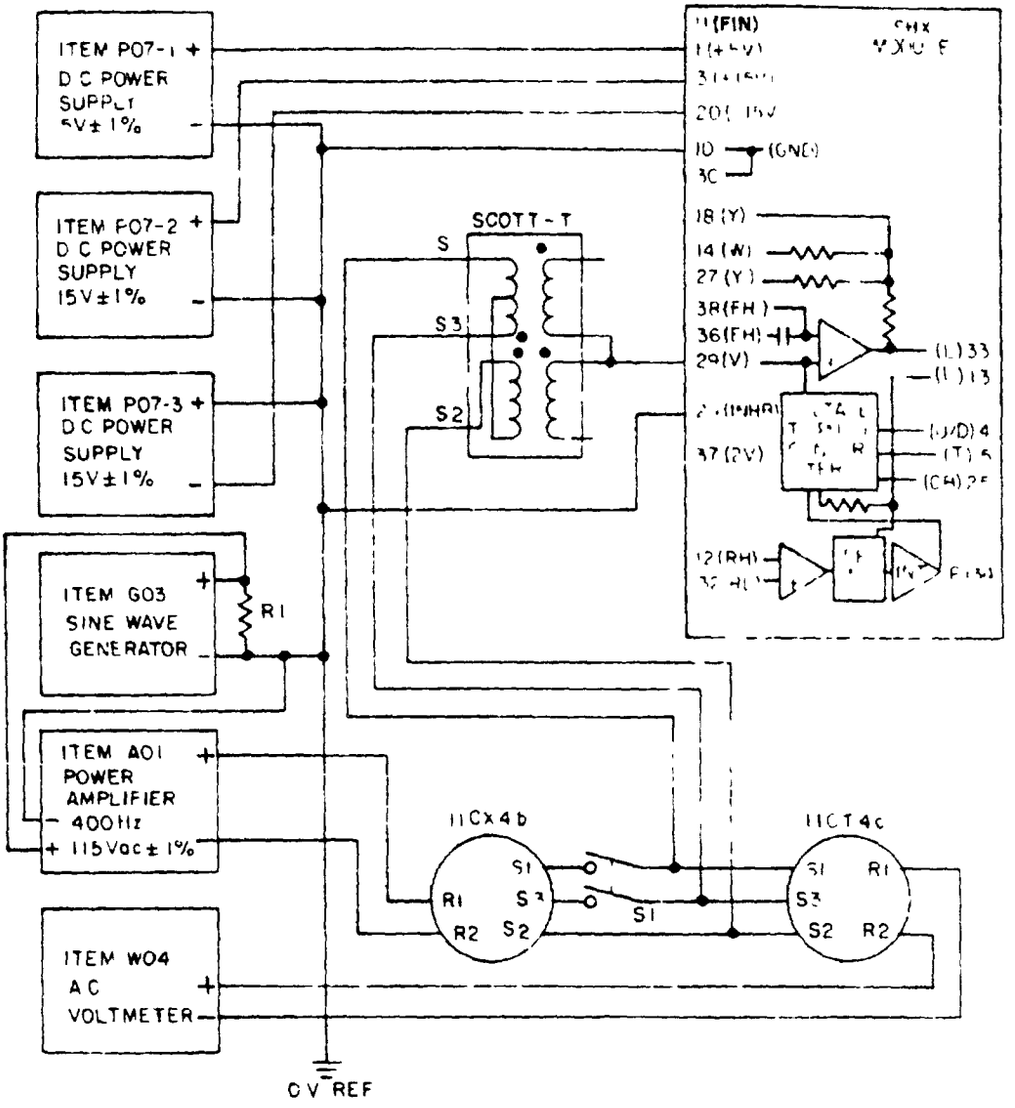
3 Load for CB line pin 25



(a) $R5 = \frac{2.4V}{100\ \mu A} = 24\ k\Omega \pm 1\%,\ 1/8\ W$
(b) CR3 = 1N4150, or equivalent
(c) Select R6 such that I3 = 3.2 mA ± 1% when point C is held at 0.4 V
(d) C3 = 100 pF ± 5%

FIGURE 7. Loads for digital outputs

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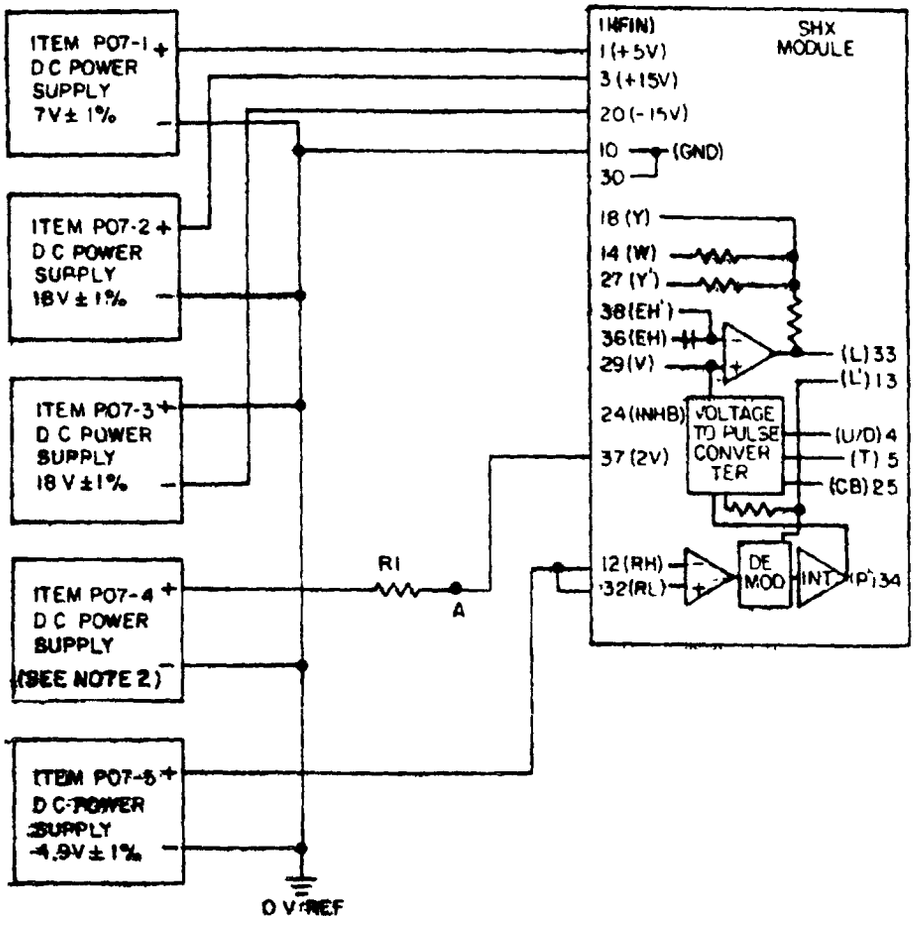


WARNING When using the A01 amplifier high voltage may be present at the output terminals Always set the voltage to zero before connecting or disconnecting the test leads

NOTES
1 R1 = 600 Ω - 5%, 1W
2 S1 - DPST switch
3 Synchros - 11CX4b and 11CT4C
4 Scott-T transformer - 90V L-L synchro signal in 1V resolver signal out
5 Test sequence
(a) Close switch S1
(b) Adjust the 11CT4c for a maximum voltage reading on item W04
(c) Open switch S1

FIGURE 8 V stress test circuit for table V

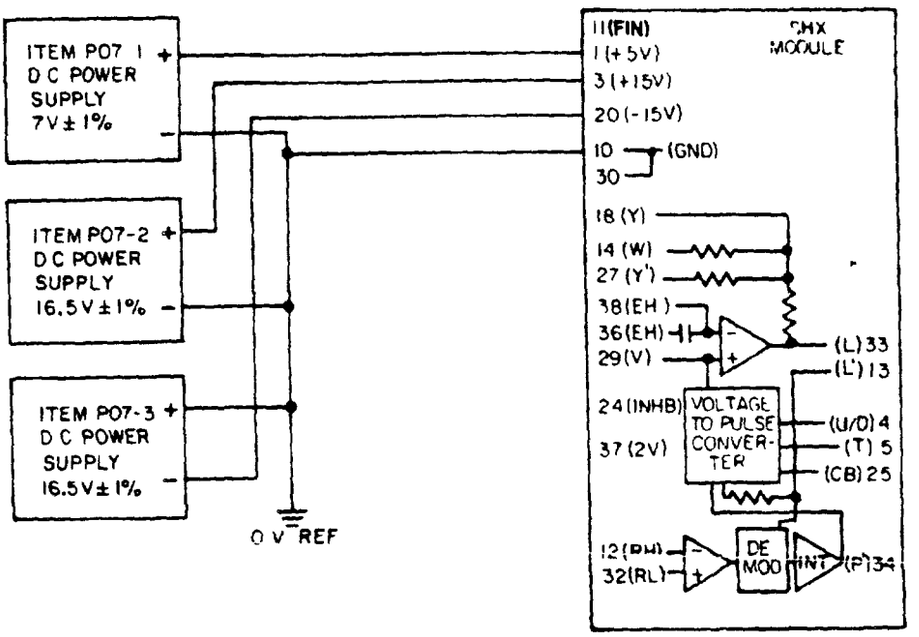
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NOTES:
1. R1 = 240Ω ±1%, 1/8-W.
2. Adjust item P07-4 for 6.8 V ±1% at point A.
3. Leave circuit connected for a minimum of 10 seconds.

FIGURE 9 Vs stress test circuit for table VI.

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NOTE Leave circuit connected for a minimum of 15 minutes

FIGURE 10. Module stress test circuit for table VII

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TABLE VIII Output short circuit current test

Test condition	Parameter	Position S1	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
G1	Output short circuit current	1	4	Open	← Survival →			11
G2			5	Open	← Survival →			
G3			23	Open	← Survival →			
G4			Open	13	15	30	mVrms	

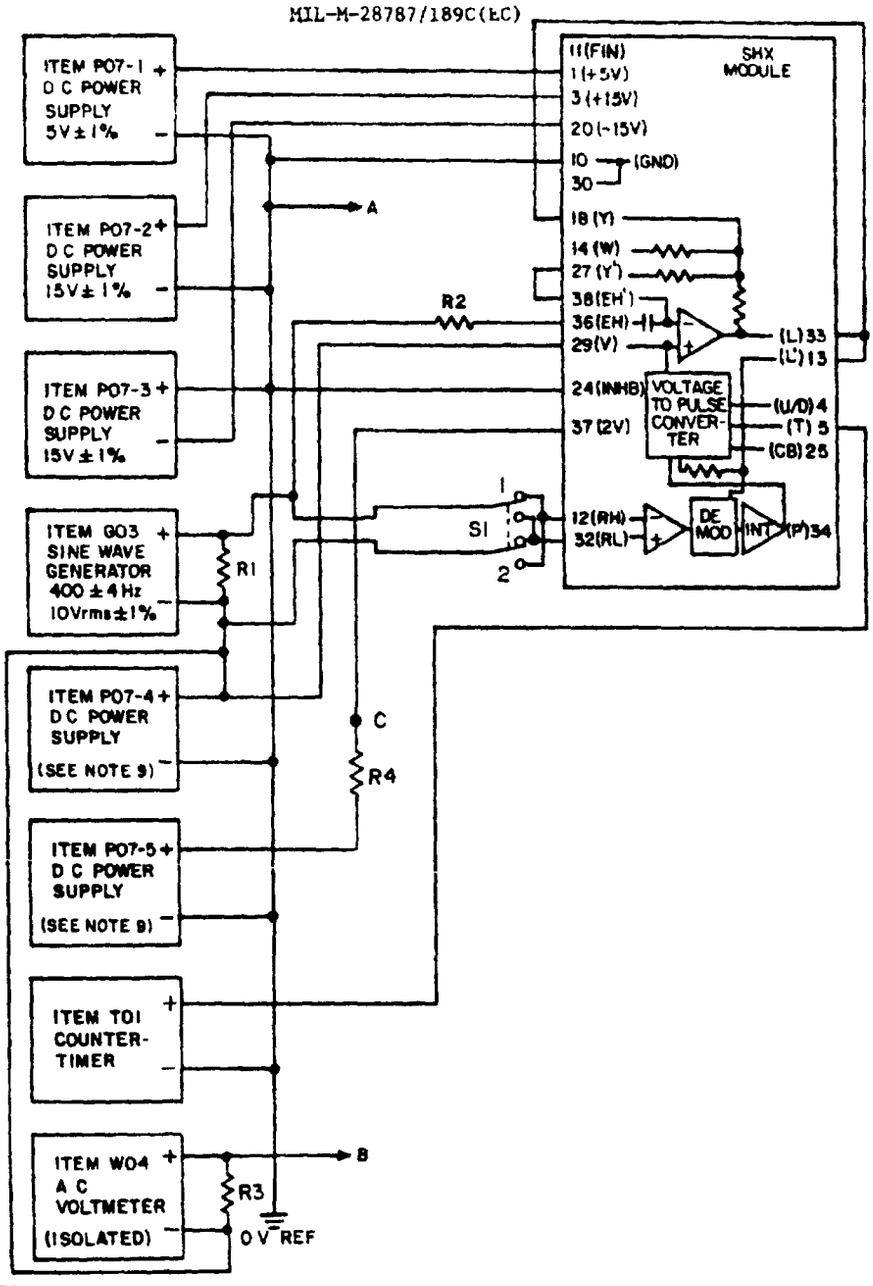
TABLE IX Maximum count test.

Test condition	Parameter	Position S1	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
H1	Maximum count	1	Open	Open	64	250	kHz	11
H2	Maximum count	2			64			
H3	Maximum count (V 4.88 V)	1			60			
H4	Maximum count (2 V 5.83 V)	2			60			

TABLE X. Scaling resistance and continuity test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
J1	Scaling resistance	14	Open	Open	3 580	3.720	kΩ	12
J2	Scaling resistance	27	Open	Open	98	102	kΩ	
J3	Scaling resistance	33	Open	Open	10.78	11 22	kΩ	
J4	Continuity (pin 11 to fin)	Open	11	Fin	0	1	Ω	
J5	Continuity (pin 10 to pin 30)	Open	10	30	0.0	0.1	Ω	
J6	Continuity (pin 10 to pin 30)	Open	10	30	0	1	Ω	

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NOTES
1. R1 = 600Ω ±10%, 1/8 W
2. R2 = 100 kΩ ±1%, 1/8 W.
3. R3 = 1Ω ±1%, 1/8 W, noninductive
4. S1 = DPDT switch.
5. R4 = 240Ω ±1%, 1/8 W
6. Load pins 4, 5, and 25 as specified on figure 7.
7. Wait 2 5 ms after S1 is in position before gatting item T01 on.
8. For test conditions G1-G3 leave module connected for a minimum of 5 seconds.
9. Adjust items P07-4 and P07-5 at point C, as follows

Test condition	P07-4	P07-5
G1, G2	4 6 V ±0.2%	6 2 V ±0 2%
G3, G4	4 880 ±0 025 V	5 830 ±0 030 V
H1, H2	4 6 V ±0 2%	6 2 V ±0 2%
H3	4 88 V ±0 2%	6 2 V ±0 2%
H4	4 6 V ±0.2%	5 83 V ±0 2%

* FIGURE 11 Output short circuit current and maximum count test circuit for tables VIII and IX

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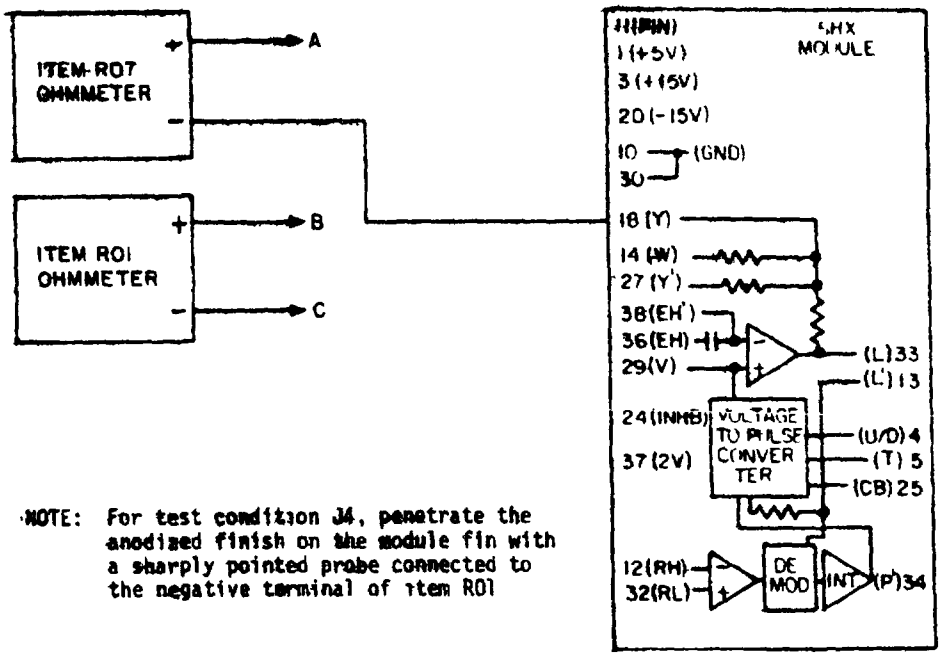


FIGURE 12. Scaling resistance and continuity test circuit for table X.

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TABLE X. Capacitance and power supply capacitance test

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Capacitance	38	36	11.25	18.75	µF	13
K2	Power supply capacitance (+15 V)	10	3	0.075	0.125		
K3	Power supply capacitance (-15 V)	20	10	0.075	0.125		
K4	Power supply capacitance (+5 V)	10	1	0.075	0.125		

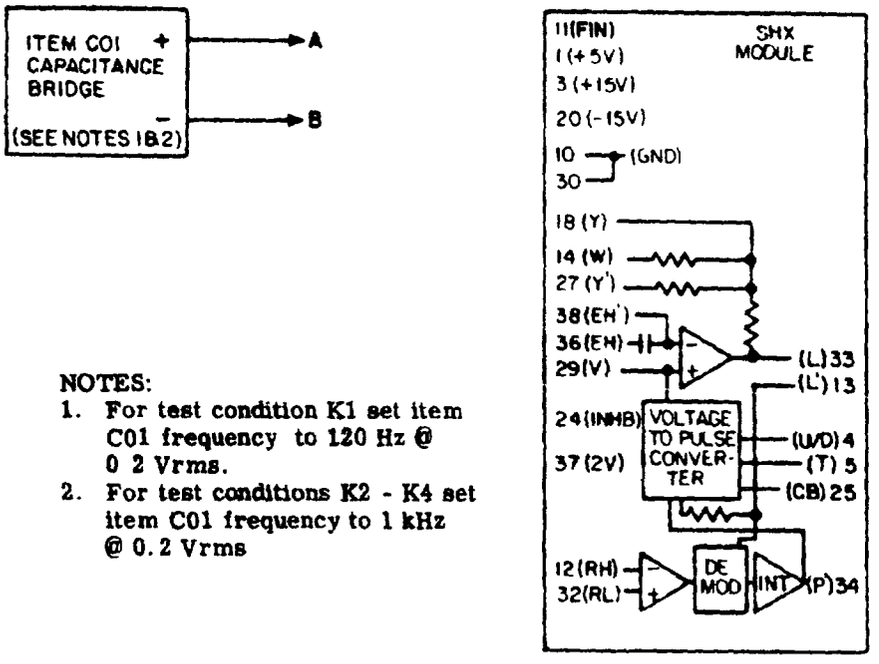
TABLE XII Input current (digital) test

Test condition	Parameter	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
L1	Input current (digital), low level	Open	24	1.0	1.4	mA	14
L2	Input current (digital), high level	24	Open	300	440	µA	14

TABLE XIII. Input current (analog) test.

Test condition	Parameter	Position S1	Connect		Initial limits 25° ±5°C Ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
M1	Input current (analog), V	1	Open	Open	-1.0	1.5	mA	15
M2	Input current (analog), V	2	Open	Open	-1.0	1.5	mA	
M3	Input current (analog), 2 V	1	Open	Open	0	1	mA	
M4	Transfer function (opamp)	1	C	33	4.80	5.20	Vrms	
M5	DC offset	1	Open	33	-25	25	mV	

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NOTES:
1. For test condition K1 set item C01 frequency to 120 Hz @ 0 2 Vrms.
2. For test conditions K2 - K4 set item C01 frequency to 1 kHz @ 0.2 Vrms

FIGURE 13 Capacitance and power supply capacitance for table XI.

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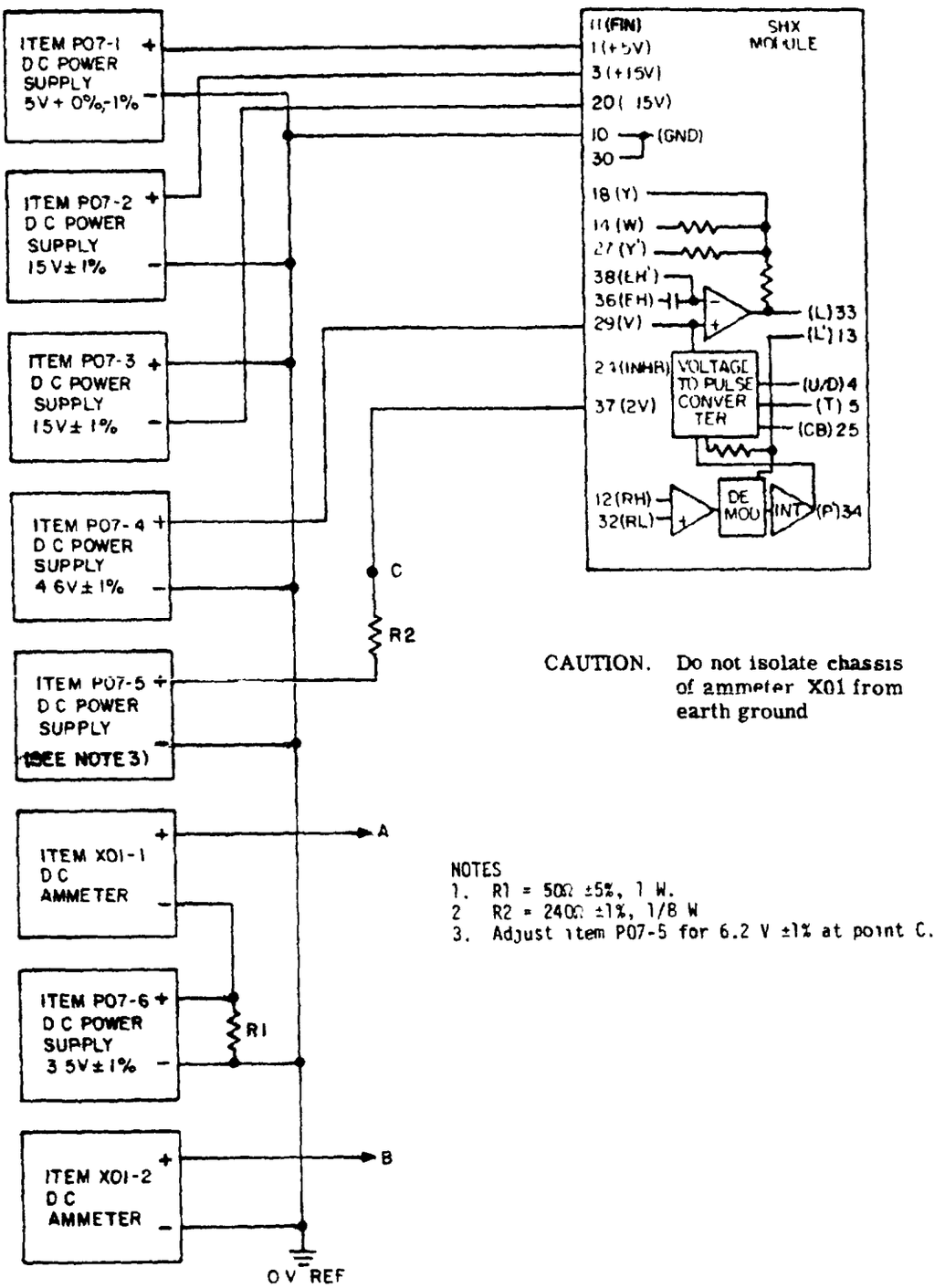
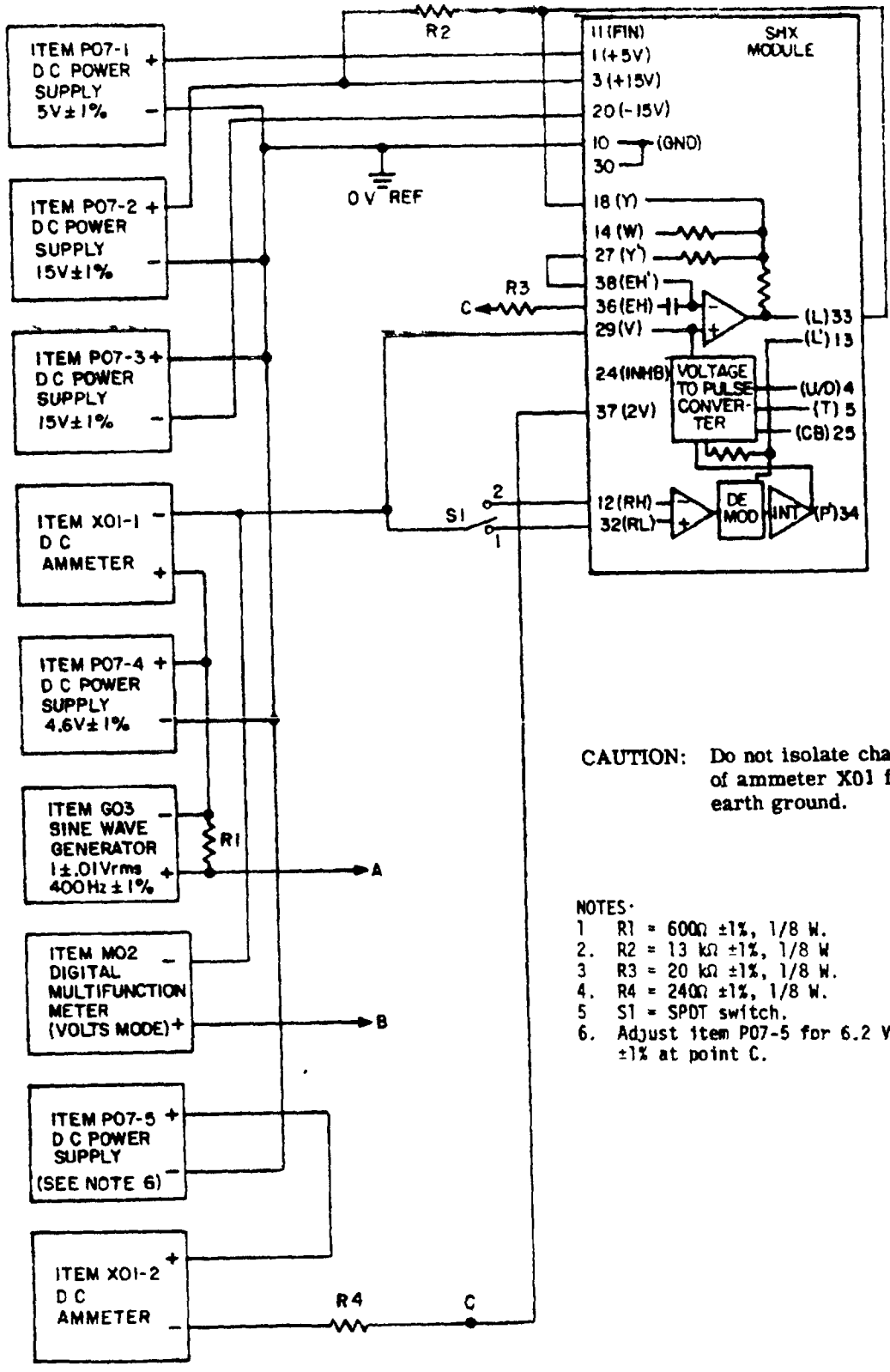


FIGURE 14 Input current (digital) test circuit for table XII

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CAUTION: Do not isolate chassis of ammeter X01 from earth ground.

NOTES:
1. R1 = 600Ω ±1%, 1/8 W.
2. R2 = 13 kΩ ±1%, 1/8 W
3. R3 = 20 kΩ ±1%, 1/8 W.
4. R4 = 240Ω ±1%, 1/8 W.
5. S1 = SPDT switch.
6. Adjust item P07-5 for 6.2 V ±1% at point C.

FIGURE 15 Input current (analog) test circuit for table XIII

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* TABLE XIV Power supply current test

Test condition	Parameter	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
N1	Power supply current (+5 V)	29	0	2	mA	16
N2	Power supply current (+15 V)	29	1	20		
N3	Power supply current (-15 V)	29	0	20		
N4	Power supply current (+5 V)	Open		35		
N5	Power supply current (+15 V)	Open		10		
N6	Power supply current (-15 V)	Open		50		

TABLE XV Dynamic test

Test condition	Parameter 1/	Position switch S1	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
P1	U/D delay, preset (t1)	1	4	-∞	-800	ns	17,18
P2	U/D delay after 1 (t2)	1		600	∞		
P3	U/D delay, preset (t3)	2		-∞	-800		
P4	U/D delay after 1 (t4)	2		600	∞		
P5	Transition time (t_{TLH})	1		15	125		
P6	Transition time (t_{TLH})	2		15	125		
P7	Transition time (t_{THL})	1		5	100		
P8	Transition time (t_{THL})	2		5	100		
P9	1 pulse width (PWA), high level	1		2	16	µs	
P10	1 pulse width (PWA), high level	2		2	16		
P11	1 pulse width (PWB), low level	1		1	16		
P12	1 pulse width (PWB), low level	2		1	16		
P13	1 high level	1		2.4	5.5	V	
P14	1 high level	2		2.4	5.5		
P15	1 low level	1		0.0	0.5		
P16	1 low level	2		0	0.5		
P17	U/D high level	1		2.4	5.5		
P18	U/D low level	2		0.0	0.5		

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TABLE XV. Dynamic test - Continued.

Test condition	Parameter 1/	Position switch S1	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
P19	CB delay (T5)	1	25	-1.7	-0.35	µs	17,18
P20	CB delay (T6)	1		0.3	4.3	µs	
P21	CB transition time (t_{TLH})	1		15	125	ns	
P22	CB transition time (t_{THL})	1		5	45	ns	
P23	CB high level	1		2.4	5.5	V	
P24	CB low level	1		0.0	0.4	V	
P25	CB low level (T7)	1		2	∞	µs	

1/ All times are referenced to t = 0 (negative edge of T line).

TABLE XVI. Inhibit delay test.

Test condition	Parameter	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
Q1	Inhibit delay (stop D1, calculation) 1/	0 0	2.5	µs	19,20
Q2	Inhibit delay (start D2)	0.0	2.5	µs	19,20

1/ For test condition Q1, D1 calculation = PWB - D3.

TABLE XVII. Input impedance check.

Test condition	Parameter	Position switch S1	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
R1	Input impedance check	1	B	0	5	Vrms	21
R2		1	C	-20	20	mVdc	
R3		2	B	0	5	Vrms	
R4		2	C	-20	20	mVdc	

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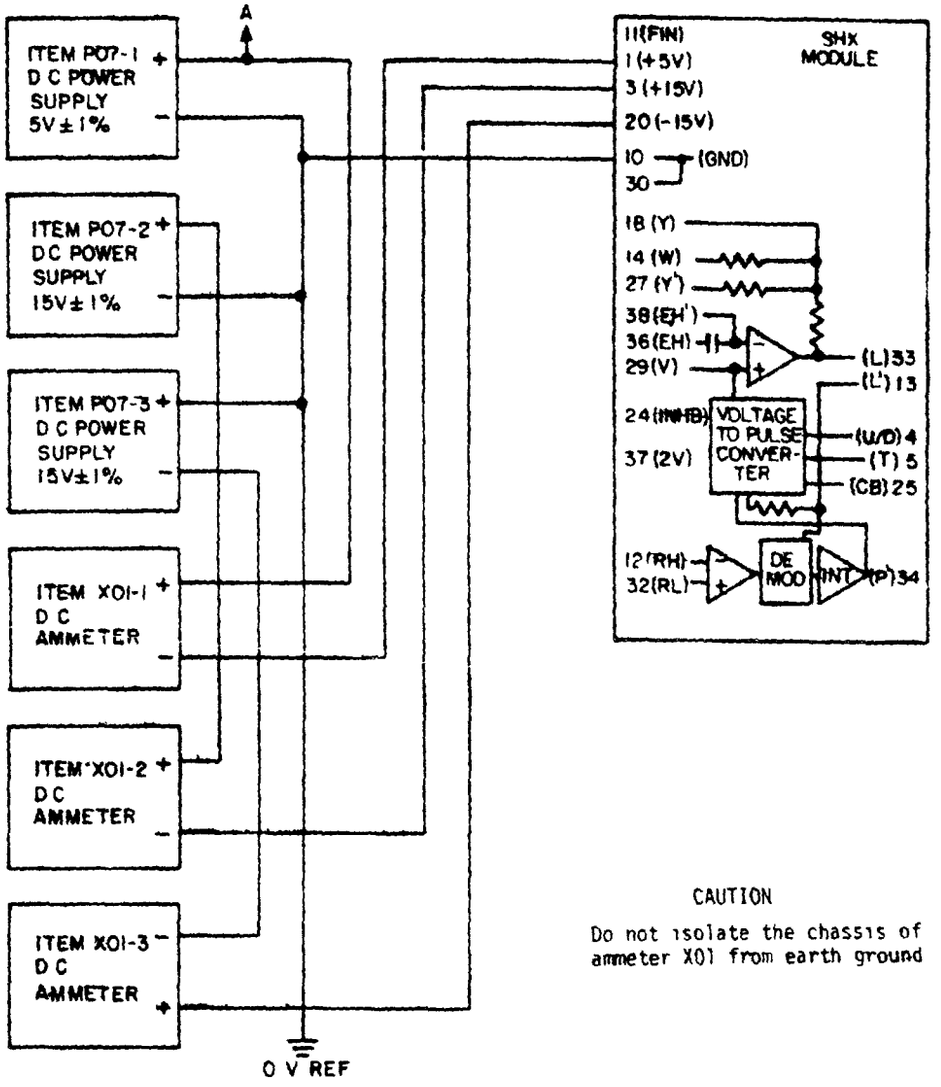
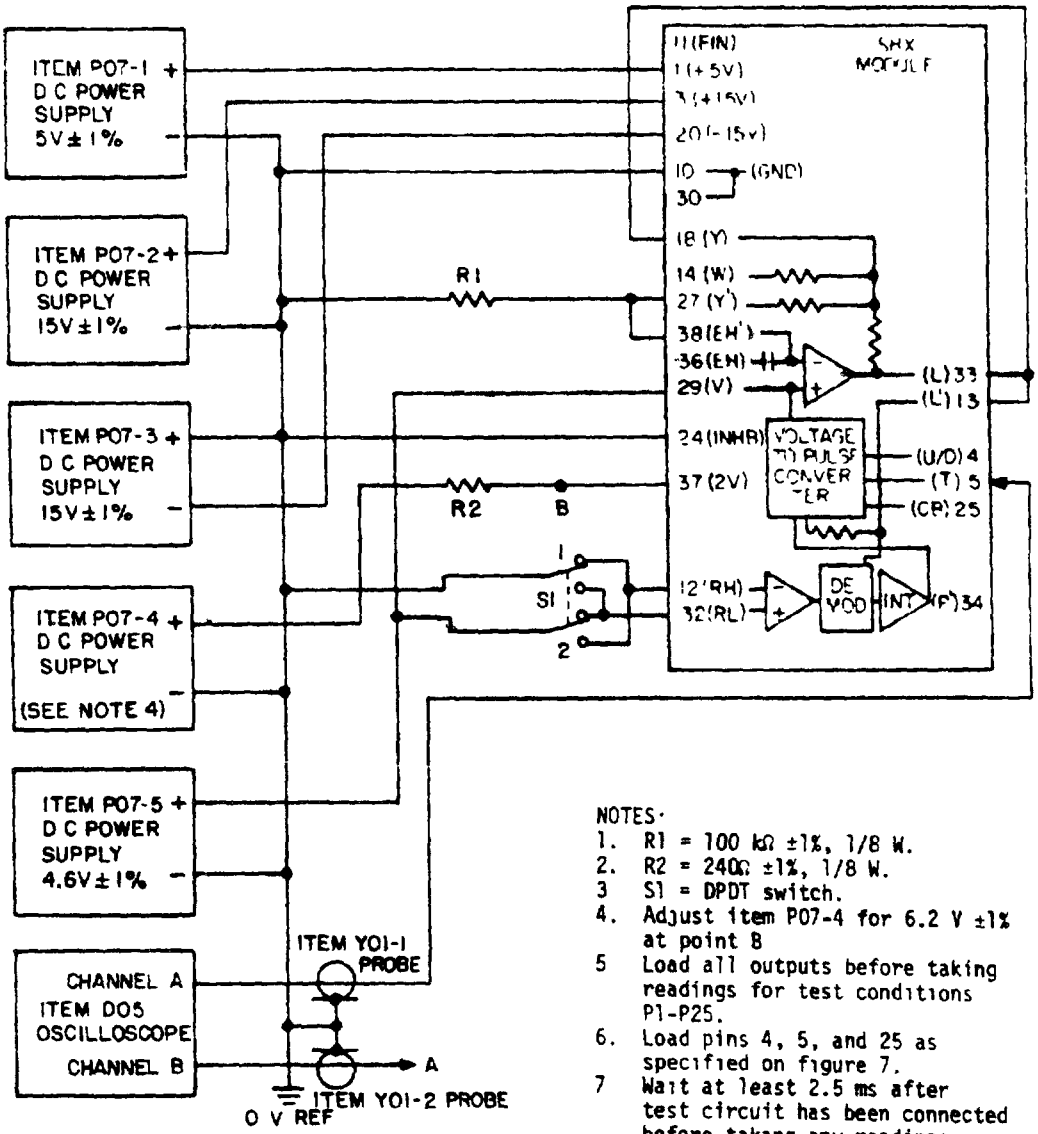


FIGURE 16. Power supply current test circuit for table XIV.

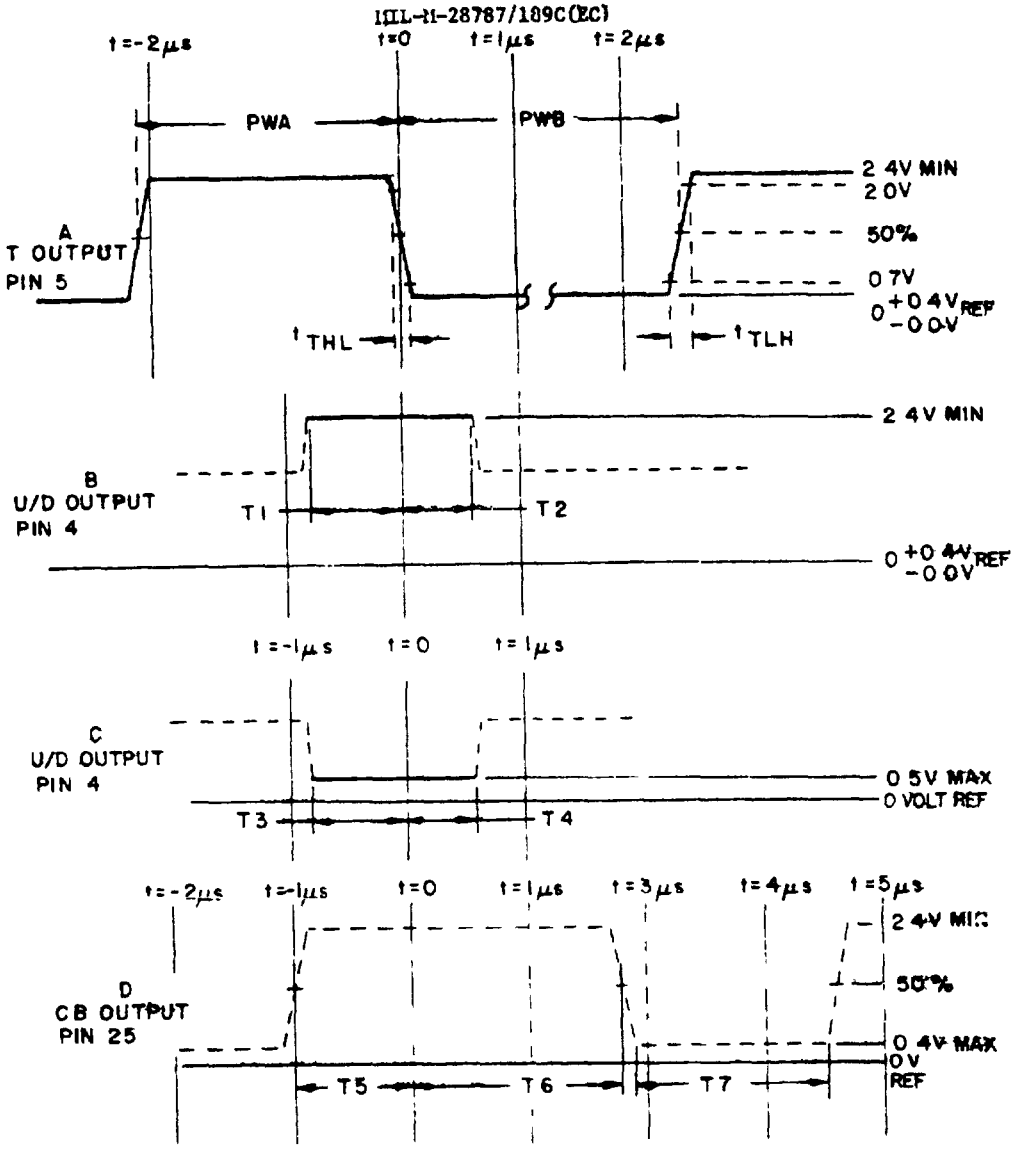
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NOTES:
1. R1 = 100 kΩ ±1%, 1/8 W.
2. R2 = 240Ω ±1%, 1/8 W.
3 S1 = DPDT switch.
4. Adjust item P07-4 for 6.2 V ±1% at point B
5 Load all outputs before taking readings for test conditions P1-P25.
6. Load pins 4, 5, and 25 as specified on figure 7.
7 Wait at least 2.5 ms after test circuit has been connected before taking any readings.

FIGURE 17 Dynamic test circuit for table XV

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NOTES

1. Waveform A is the output of pin 5 (T) with times of all waveforms referenced to the 50% high-to-low transition of waveform A

2.

Waveform	Test condition
A, B	P1, P2
A, C	P3, P4
A	P5 - P16
B	P17
C	P18
A, D	P19, P20
D	P21 - P25

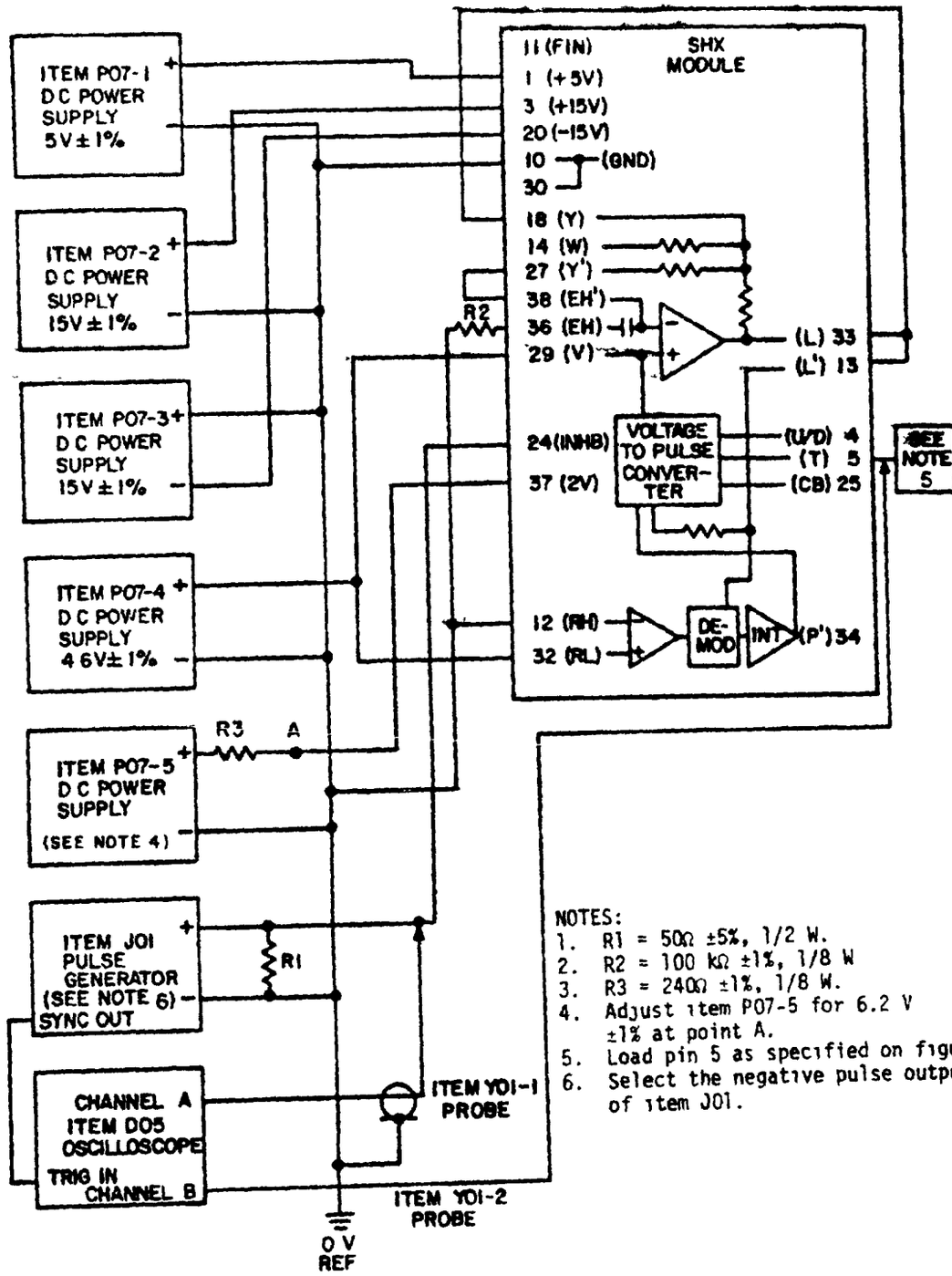
3. All transition times (t_{THL}) are from the 2 0 V level to the 0 7 level.

4. All transition times (t_{TLH}) are from the 0.7 V level to the 2.0 V level.

5. All voltage levels are referenced to zero volts.

FIGURE 18. Dynamic test circuit waveforms for table XV.

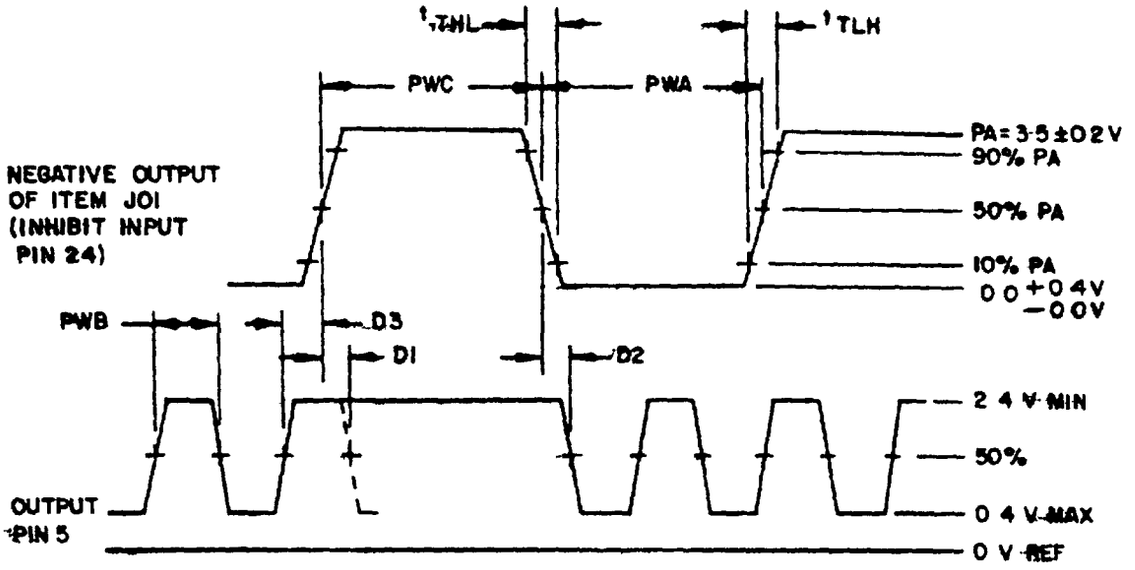
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NOTES:
1. R1 = 50Ω ±5%, 1/2 W.
2. R2 = 100 kΩ ±1%, 1/8 W
3. R3 = 240Ω ±1%, 1/8 W.
4. Adjust item P07-5 for 6.2 V ±1% at point A.
5. Load pin 5 as specified on figure 7.
6. Select the negative pulse output of item J01.

FIGURE 19. Inhibit delay test circuit for table XVI.

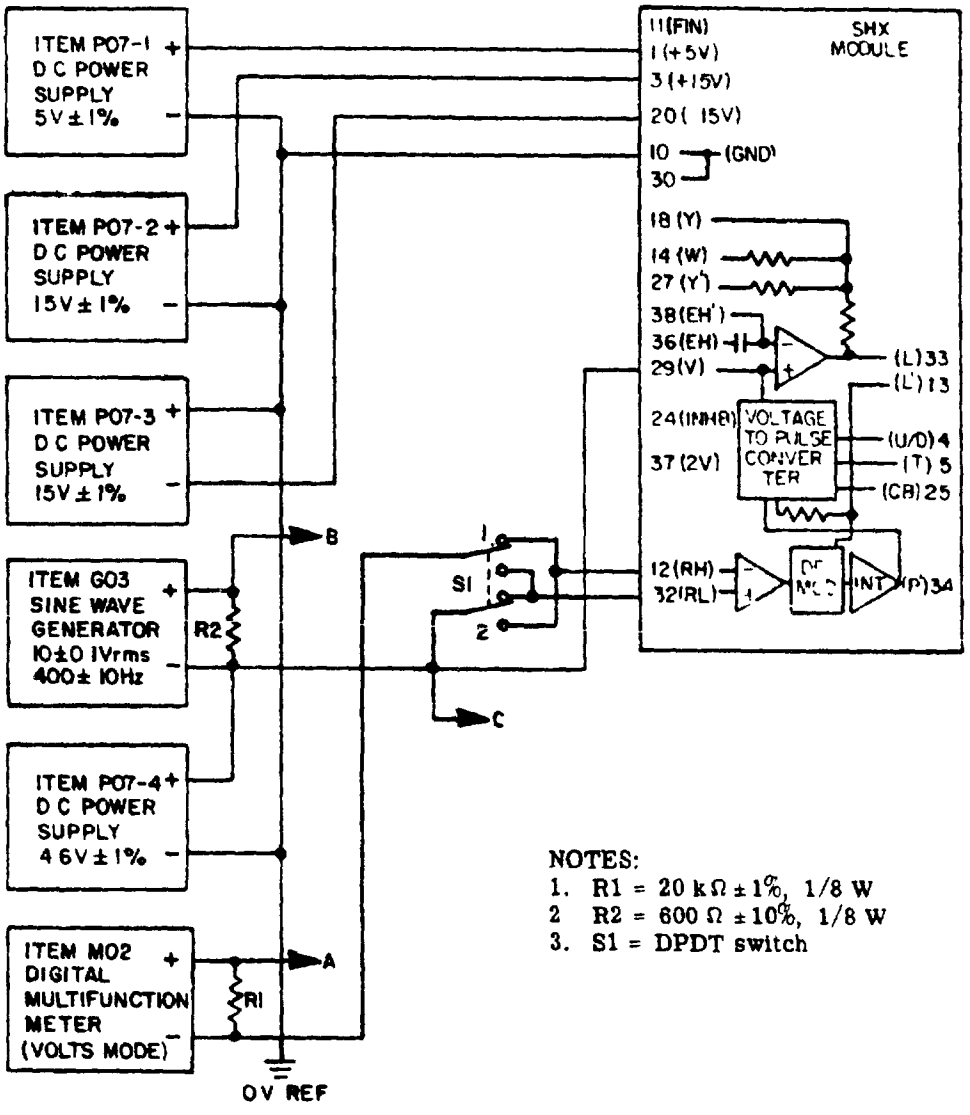
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NOTES:
1. Item J01 setup:
   a. By using the negative output of item J01, PWA can be adjusted; however, the low level must be readjusted using the baseline voltage adjustment.
   b. Initially set PWA = PWC = 50 µs ±10%.
   c. $t_{TLH} = t_{THL} = 50 \pm 10$ ns.
   d. For test condition Q1 adjust item J01 pulse width (PWA) for three negative pulses at output pin 5, then increase the PWA pulse width for a maximum D3 delay (increase PWA until a fourth negative pulse starts to appear).
2. For test condition Q1 the inhibit delay (stop D1) is equal to the measured PWB minus the maximum measured D3 reading.

FIGURE 20. Inhibit delay test circuit waveforms for table XVI.

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NOTES:
1. R1 = 20 kΩ ± 1%, 1/8 W
2 R2 = 600 Ω ± 10%, 1/8 W
3. S1 = DPDT switch

FIGURE 21 Input impedance check test circuit for table XVII.

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TABLE XVIII Gain test

Test condition	Parameter	Position switch S1	Connect A to pin	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
S1	Gain	1	34	4	6	Vp-p	22
S2		1		4	6		
S3		1		3.0	4 5		
S4		2		4	6		
S5		2		4	6		
S6		2		3 0	4 5		

TABLE XIX. Voltage controlled oscillator test

Test condition	Parameter	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
F1	Voltage controlled oscillator (2 V)	4	40	kHz	23
F2	Voltage controlled oscillator (5 V)	10	105		
F3	Voltage controlled oscillator (2 V)	4	40		
F4	Voltage controlled oscillator (5 V)	10	105		

TABLE XX. Null zone test

Test condition	Parameter	Position switch S2	Initial limits 25° ±5°C Ambient Min	Initial limits 25° ±5°C Ambient Max	Unit	Test figure
U1	Null zone (R3 + R4)	1	97 6	720 0	kΩ	24
U2	Null zone (R3 + R4)	2	97 6	720 0		
U3	Null zone (R3 + R4)	1	97 6	720.0		
U4	Null zone (R3 + R4)	2	97 6	720 0		
U5	Null zone calculation 1/ $\left[\frac{0.1\ V}{0\ 69\ \mu A}\left(\frac{1}{U1}+\frac{1}{U2}\right)\right]$	--	1 3	3 0	--	

1/ Ensures that a synchro could be moved through a 1 3 least significant bit arc and there would not be an output on pin 5

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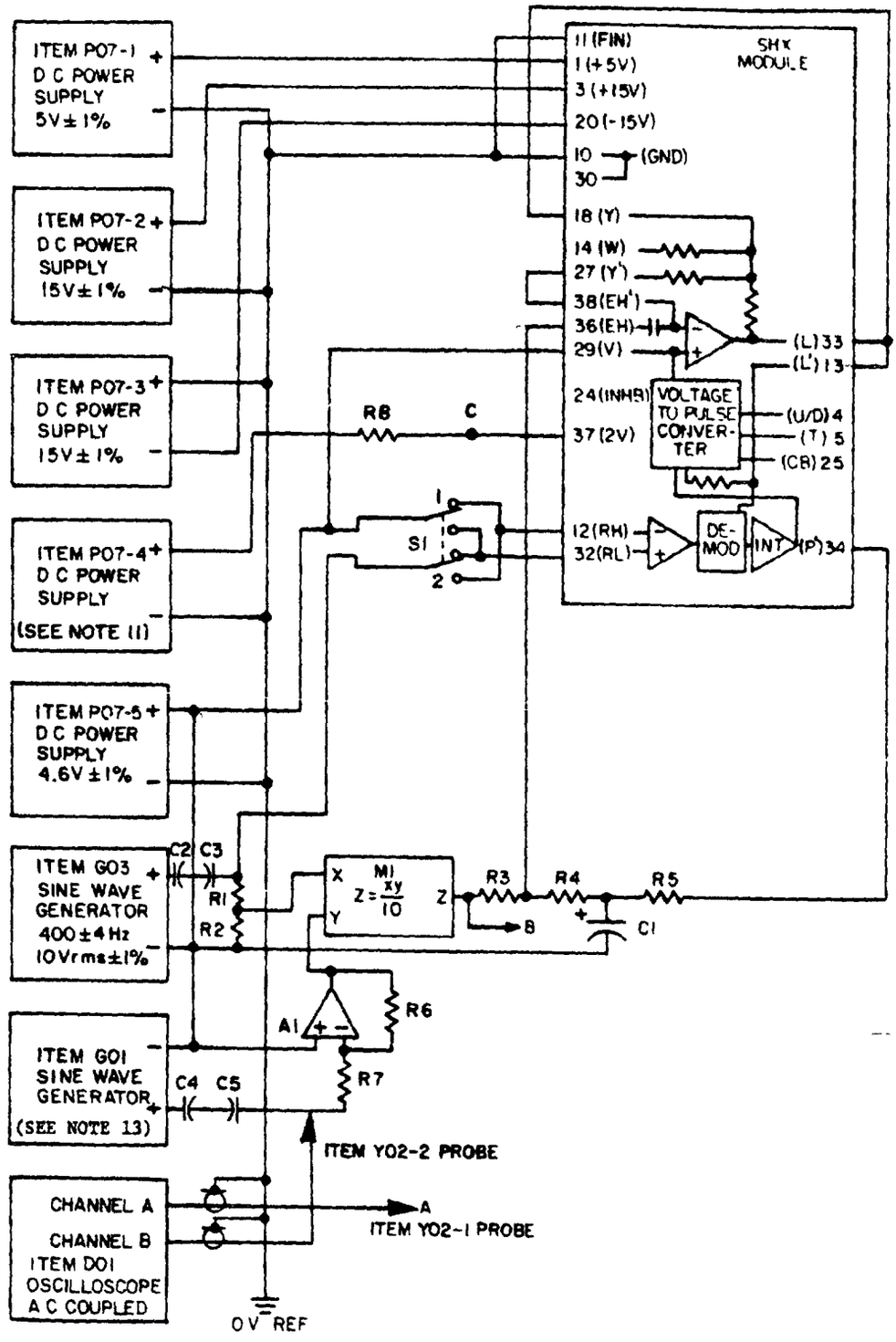


FIGURE 22 Gain test circuit for table XVIII

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NOTES
1. R1 = R2 = 1 kΩ ±1%, 1/8 W.
2. R3 = 110 kΩ ±0.1%, 1/8 W.
3 R4 = R5 = 20 kΩ ±0 1%, 1/8 W.
4. R6 = R7 = 10 kΩ ±1%, 1/8 W.
5 R8 = 240Ω ±1%, 1/8 W.
6. C1 = 1,000 µF ±10%, 6 Vdc, electrolytic.
7. C2-C5 = 200 µF ±20%, 25 Vdc
8. S1 = DPDT switch.
9 A1 = Fairchild µA 741, or equivalent
10. M1 = AD530LD, or equivalent
11 Adjust item P07-4 for 6.2 V ±1% at point C.
12. $Z = \frac{xy}{10}$
13. Connect point A to point B, adjust item G01 for 1 0 V_{p-p} ±3%, 1.6 Hz ±1%, at the output of M1 for test conditions S1 and S4; adjust item G01 for 5.0 V_{p-p} ±3%, 64 Hz ±1%, at the output of M1 for test conditions S2 and S5, adjust item G01 for 5.0 V_{p-p} ±3%, 160 Hz ±1%, at the output of M1 for test conditions S3 and S6
14 Power supply returns for A1 and M1 shall be connected to analog ground

FIGURE 22 Gain test circuit for table XVIII - Continued.

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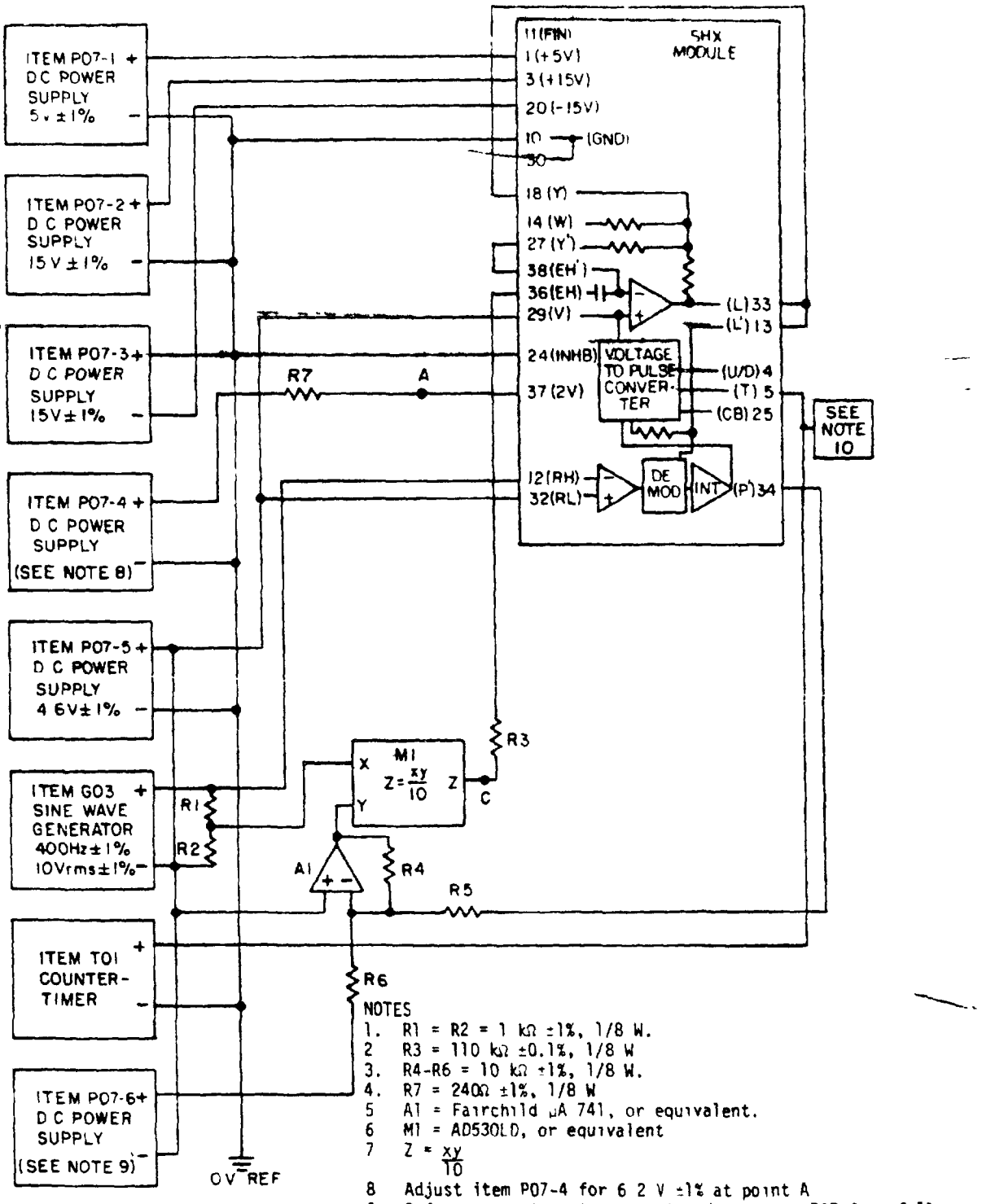


FIGURE 23 Voltage controlled oscillator test circuit for table XIX

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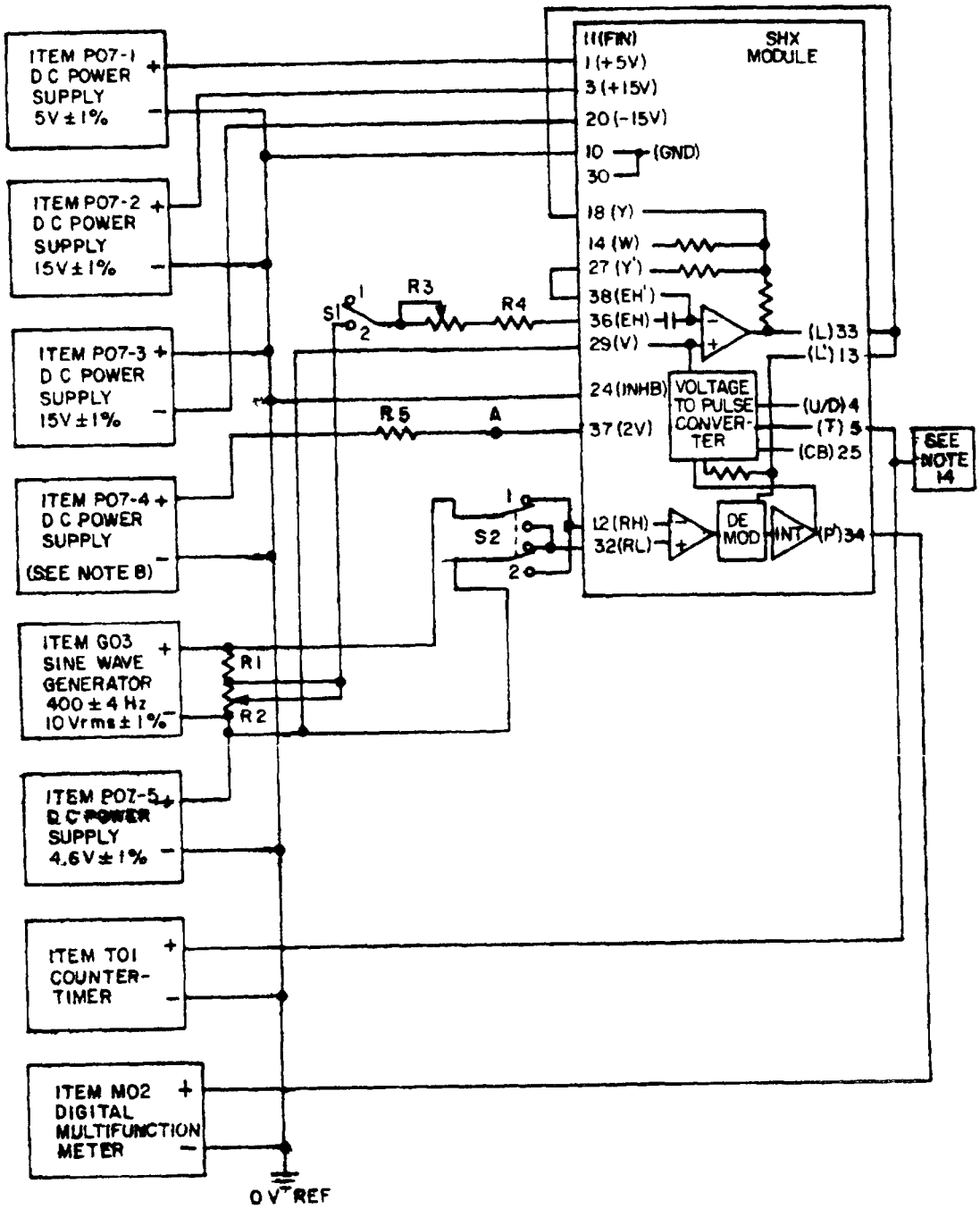


FIGURE 24 Null zone test circuit for table XX

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NOTES:
1. R1 = 1 kΩ ±1%, 1/8 W.
2. R2 = 15Ω variable, 1/8 W.
3. R3 = 750 kΩ variable, 1/8 W.
4. R4 = 97.6 kΩ ±1%, 1/8 W.
5. R5 = 240Ω ±1%, 1/8 W.
6. S1 = SPDT switch.
7. S2 = DPDT switch.
8. Adjust item P07-4 for 6.2 V ±1% at point A.
9. Set R2 so that the voltage across it is 0.1 Vrms ±1%.
10. Before S1 is placed in position 2, ensure pin 34 is at analog ground and R3 has the maximum value.
11. For test conditions U1 and U2 adjust R3 very slowly until pin 5 begins to toggle (*). Open S1 and back off R3 a little. Before S1 is closed-ensure that pin 34 is at analog ground Pin 5 should not toggle If pin 5 does toggle, open S1 and back off R3 a little more Repeat this until pin 5 does not toggle for a minimum of 2 seconds after S1 is closed.
12. Measure series resistance of R3 and R4 combination for test conditions U1 through U4.
13. For test conditions U3 and U4, pin 5 should toggle.
14. Load pin 5 output as specified on figure 7.
15. For test condition U5 the number to be compared to the limits is calculated from the following formula:

$$\frac{0.1\ V}{0.69\ \mu A}\left(\frac{1}{U1}+\frac{1}{U2}\right)$$

* Alternates between two states.

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FIGURE 24 Null zone test circuit for table XX - Continued.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-28787/189C(EC) Modules, Standard Electronic Processor, Error, S/D, Analog, Key Code SHX

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